

Lab 1: Requirements Engineering & UML Use-Case Modelling

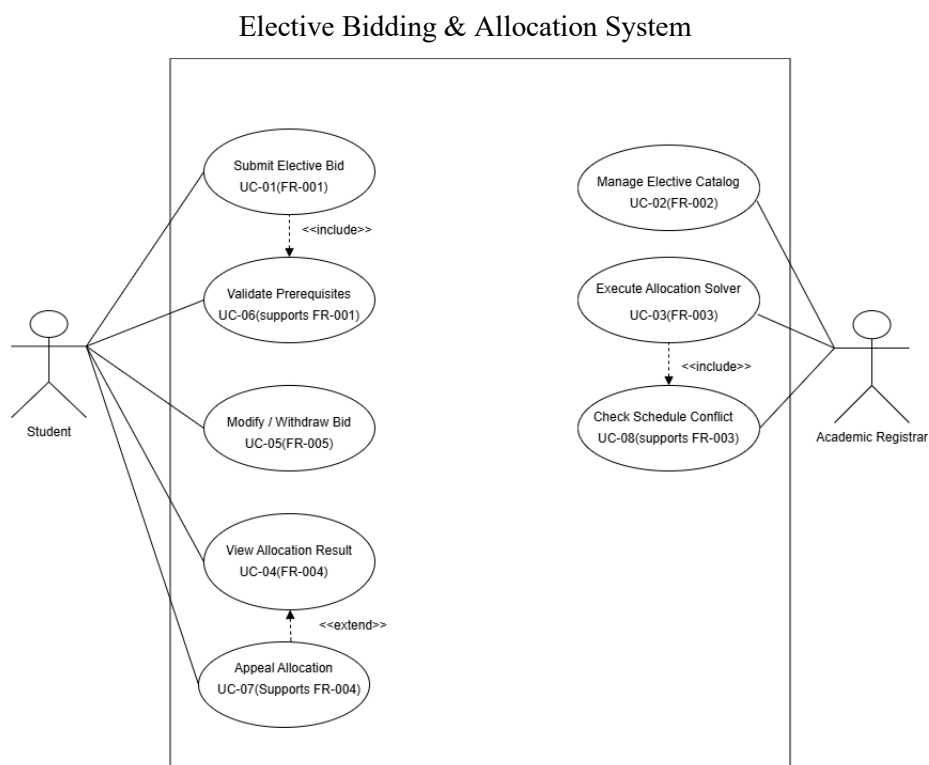
Team_5: Academic Elective Bidding & Allocation System

Requirements Table:

Row highlighted in BLUE (FR-003) is the requirement owned by Me(PESIUG24AM906_PallaviB)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core bidding mechanism; ensures fair credit-based ranking and academic eligibility. (Given)
FR-002	Functional	The system shall allow the Academic Registrar to configure the elective course catalog, including course list, prerequisites, seat capacity, and timetable slots for the term.	High	Pass: All configured courses appear correctly in the bidding interface with accurate seats/slots. Fail: Any course missing capacity or slot data.	Bidding and allocation cannot begin without a correctly configured catalog.
FR-003	Functional	The system shall allow the Academic Registrar to execute the allocation solver, which assigns electives to students based on submitted bids, seat capacity, and prerequisite eligibility while eliminating timetable clashes.	High	Pass: Output allocation has zero students with overlapping-slot electives and zero courses exceeding seat capacity. Fail: Any conflict or overcapacity found in the output.	Core engine functionality that fulfils the fair, constraint-respecting allocation promised to students.
FR-004	Functional	The system shall allow students to view their final elective allocation and resulting timetable once the allocation solver has completed a run.	High	Pass: Allocated course(s) and schedule display accurately within 2 seconds of solver completion. Fail: Result missing, delayed, or shows unallocated/incorrect data.	Students need timely, accurate visibility into their outcome to plan their semester.
FR-005	Functional	The system shall allow students to withdraw or modify a submitted bid at any time before the bidding window closes.	Medium	Pass: Updated bid reflects the new credit distribution and previous bid is fully overwritten. Fail: Edit accepted after deadline or partial update applied.	Gives students flexibility to correct mistakes without registrar intervention, up to a firm cut-off.
NFR-001	Nonfunctional	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the allocation engine scales to full-university peak load without delaying result publication. (Given)
NFR-002	Nonfunctional	The system shall log every allocation solver run (timestamp, registrar ID, input bid snapshot, output allocation) and restrict solver execution to users with Registrar privileges.	Medium	Pass: Unauthorized (Student) execution attempts are rejected; every run produces an immutable audit record. Fail: Solver runs without logging or is triggerable by a non-Registrar account.	Supports academic-integrity audits and dispute resolution when students appeal their allocation.

UML Use-Case Diagram: Academic Elective Bidding & Allocation System:



Use-Case Flow Specification:

Field	Details
Use Case	Execute Allocation Solver (UC-03 / FR-003)
Primary Actor	Academic Registrar
Secondary Actor	Student (indirect beneficiary — receives allocation output)
Includes	Check Schedule Conflict «include» — always performed as part of this use case
Trigger	Registrar closes the elective bidding window and initiates the allocation run for the term.

Preconditions

1. The bidding window has closed and all student bids have been submitted (Use Case: Submit Elective Bid is complete for the term).
2. The elective course catalog — course list, seat capacity, and timetable slots — has been configured (Use Case: Manage Elective Catalog is complete).
3. The Academic Registrar is authenticated and holds allocation-execution privileges.

Postconditions

Success:

Every student bid has been processed into a final allocation (assigned elective or waitlisted) with no timetable collisions and no course exceeding seat capacity. Students are notified that results are available. The run is logged for audit.

Failure:

No partial allocation is committed. The system reverts to the pre-run state, logs the failure reason, and notifies the Registrar.

Main Success Scenario

1. Registrar logs in and opens the Allocation Dashboard.
2. Registrar selects the current bidding cycle/term and selects "Run Allocation Solver."
3. System retrieves all submitted student bids together with their validated prerequisite status.
4. System runs the allocation optimization, matching each student's ranked, credit-weighted preferences to available elective seats within capacity limits.
5. System checks the proposed allocation for timetable collisions across each student's assigned electives («include»: Check Schedule Conflict).
6. System finalizes and stores the allocation results.
7. System notifies all students that results are available and records the run (timestamp, Registrar ID, bid snapshot, output) in the audit log.
8. Registrar reviews the summary report (seats filled, unmet bids, conflicts resolved) and confirms publication.
9. Use case ends successfully.

Alternate Flow — 5a: Timetable Conflict Cannot Be Auto-Resolved

- 5a1. During conflict checking, the system finds a student with two mandatory electives that share an identical timeslot with no valid substitute.
- 5a2. System excludes the affected student record(s) from the auto-committed allocation and generates an exception report.
- 5a3. Registrar manually reassigns or defers the affected student(s) to a supplementary allocation round.
- 5a4. Flow resumes at Main Success Scenario step 6 with the corrected allocation set.